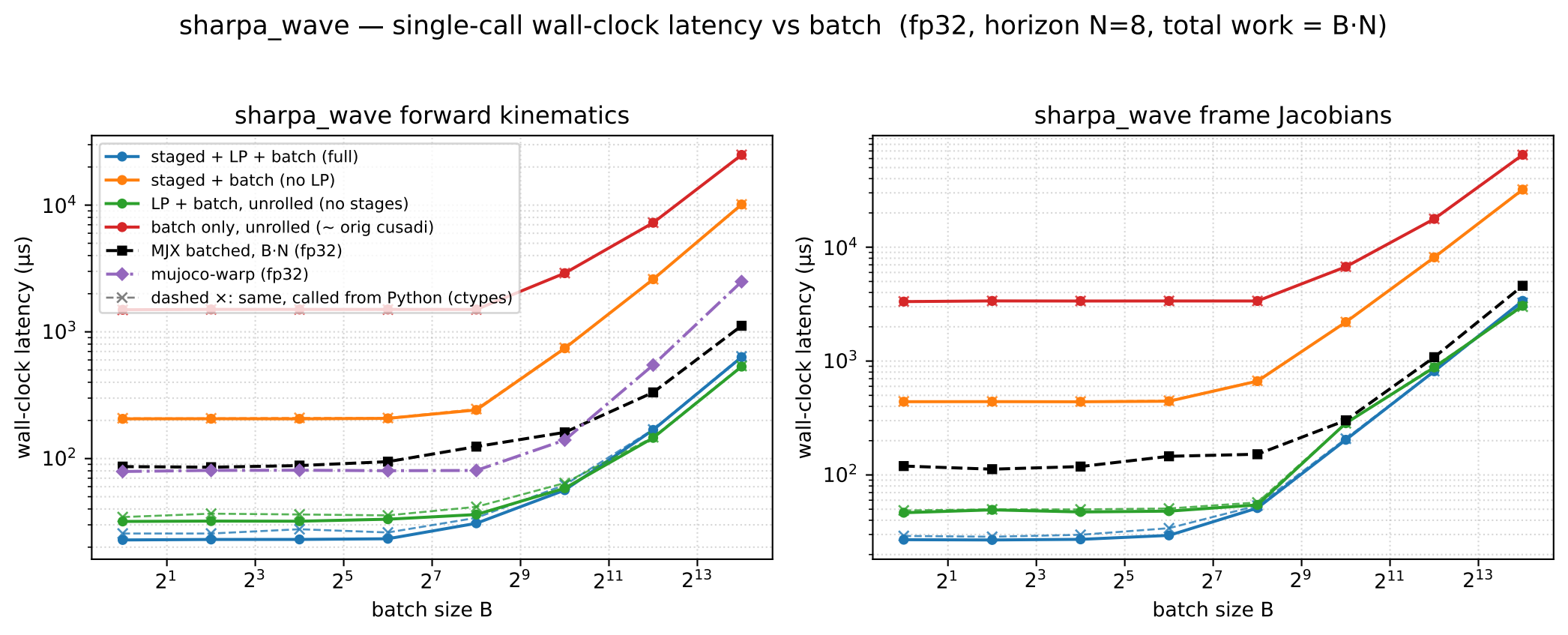
sharpa_wave — single-call wall-clock latency vs batch (fp32, horizon N=8, total work = B·N)
sharpa_wave forward kinematics
sharpa_wave frame Jacobians
staged + LP + batch (full)
staged + batch (no LP)
LP + batch, unrolled (no stages)
batch only, unrolled (~ orig cusadi)
MJX batched, B·N (fp32)
mujoco-warp (fp32)
dashed ×: same, called from Python (ctypes)
wall-clock latency (µs)
batch size B